Oindrila Sinha, Ph.D.

Postdoctoral Fellow · Georgia Institute of Technology and Emory School of Medicine

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Research Interests

How perceptual learning becomes motor skill; how this relationship varies across adulthood; and how cognitive abilities contribute to individual differences in learning. My research combines anticipatory gaze, motor behavior, EEG, wearable sensing, and machine learning to explain why people respond differently to the same practice.

Methods: Eye tracking, EEG, motor-behavior assessment, neuromodulation, statistics, machine learning, computational modeling, and artificial neural networks.

Education

Ph.D. in Motor Control Aug 2020–Dec 2024
The Pennsylvania State University · Department of Kinesiology
Minor in Statistics · Department of Statistics

Master of Science Aug 2018–May 2020
Presidency University · Department of Life Sciences

Bachelor of Science Aug 2015–May 2018
Presidency University · Department of Life Sciences

Research Experience

Postdoctoral Fellow Jan 2025–Present
Georgia Institute of Technology and Emory School of Medicine

Develop protocols combining gaze, motor responses, and EEG to identify markers of motor learning. Apply embedded sensing and machine learning to characterize functional prosthesis engagement.

Doctoral Student and Teaching Assistant Aug 2020–Dec 2024
The Pennsylvania State University

Investigated visual contributions to anticipatory hand forces, postural preparation, and rapid corrective responses during interactions with moving objects.

Project Research Scientist Aug 2018–Aug 2020
S. N. Bose National Centre for Basic Sciences

Developed biomedical and environmental sensing systems using optics, EEG, motion tracking, and signal analysis; contributed to biofilm-treatment research.

Project Research Scientist May–Aug 2019
Indian Institute of Technology Gandhinagar

Developed a wearable fiber Bragg grating system to detect pre-dicrotic and dicrotic notches in arterial-pulse waveforms.

Project Research Scientist May 2016–May 2018
Bose Institute

Developed an ultrasound navigation aid for people with visual impairments and an EEG/eye-blink drowsy-driver detection system that contributed to a patent.

GRANT PROPOSALS

Submitted and pending

PI · American Academy of Neurology Sep 2026

Aging effects on predictive cognition during sequence learning.

Co-PI · NSF DARE Jun 2026

Soft sensor technologies for real-time assessment of upper-limb prosthesis function.

Co-PI · NIH R01 Feb 2026

Sensor-based measures of upper-limb prosthesis wear and functional use.

PUBLICATIONS AND PREPRINTS

10 peer-reviewed journal articles · 3 preprints · 2 conference publications

Preprints

15. **Sinha, O.**, Crews, L. G., Pleasant, W. A., Borich, M. R., & Wheaton, L. A. (2026). Anticipatory gaze reveals individual differences in implicit sequence learning before motor performance. bioRxiv [Preprint]. https://doi.org/10.64898/2026.09.09.750154

14. **Sinha, O.**, Kurtzer, I., & Singh, T. (2026). Aging selectively impairs how peripheral vision calibrates anticipatory postural responses to object motion. bioRxiv [Preprint]. https://doi.org/10.64898/2026.05.07.723563

13. Singh, S., Halder, A., Banerjee, A., Hasan, Md. N., Bera, A., **Sinha, O.**, Ghosh, S. K., Mitra, A., & Pal, S. K. (2020). An optical scattering based cost-effective approach towards quantitative assessment of turbidity and particle size estimation in drinking water using image analysis. EcoEvoRxiv [Preprint]. https://doi.org/10.32942/osf.io/davfq

Peer-Reviewed Journal Articles

12. **Sinha, O.**, Kang, T. W., Yeo, W.-H., & Wheaton, L. A. (2026). Monitoring functional engagement of upper-limb prostheses using embedded sensors and machine learning in real-world contexts. Flexible and Printed Electronics, 11, 025001. https://doi.org/10.1088/2058-8585/ae511a

11. Gutierrez, I., **Sinha, O.**, Baune, N., Mirdamadi, J., & Borich, M. (2026). Disrupting motor cortical regional activity during motor sequence skill training impairs human motor visuomotor skill acquisition and learning that is not sequence-specific. eNeuro, 13(5), ENEURO.0348-25.2026. https://doi.org/10.1523/ENEURO.0348-25.2026

10. **Sinha, O.**, Muttee, A. P., Wu, J.-H., Bertucco, M., Kurtzer, I., & Singh, T. (2025). Smooth pursuit eye movements contribute to long-latency reflex modulation in the lower extremity. Journal of Neurophysiology, 134, 998–1006. https://doi.org/10.1152/jn.00023.2025

9. **Sinha, O.**, Rosenquist, T., Fedorshak, A., Kpankpa, J., Albenze, E., Bonnet, C., Bertucco, M., Kurtzer, I., & Singh, T. (2024). Predictive posture stabilization before contact with moving objects: Equivalence of smooth pursuit tracking and peripheral vision. Journal of Neurophysiology, 132, 695–709. https://doi.org/10.1152/jn.00158.2024

8. Nardon, M., **Sinha, O.**, Kpankpa, J., Albenze, E., Bonnet, C., Bertucco, M., & Singh, T. (2024). Prioritized adjustments in posture stabilization and adaptive reaching during neuromuscular fatigue of lower-limb muscles. Journal of Applied Physiology, 137(3), 629–645. https://doi.org/10.1152/japplphysiol.00252.2024

7. **Sinha, O.**, Madarshahian, S., Gómez-Granados, A., Paine, M. L., Kurtzer, I., & Singh, T. (2023). Smooth pursuit eye movements contribute to anticipatory force control during mechanical stopping of moving objects. Journal of Neurophysiology, 129(6), 1293–1309. https://doi.org/10.1152/jn.00075.2023

6. Bhattacharyya, N., Singh, S., Banerjee, A., Ghosh, R., **Sinha, O.**, Das, N., Gayen, R., Pal, S. S., Ganguly, S., Dasgupta, T., Dasgupta, T., Mondal, P., Adhikari, A., Sarkar, S., Bhattacharyya, D., Mallick, A. K., Singh, O. P., & Pal, S. K. (2022). Integration of electroencephalogram (EEG) and motion tracking sensors for objective measure of attention-deficit hyperactivity disorder (MAHD) in pre-schoolers. Review of Scientific Instruments, 93(5), 054101. https://doi.org/10.1063/5.0088044

5. Singh, S., Halder, A., **Sinha, O.**, Chakrabarty, N., Chatterjee, T., Adhikari, A., Singh, P., Shikha, D., Ghosh, R., Banerjee, A., Das Mahapatra, P. P., Mandhar, A., Bhattacharyya, M., Bose, S., Ahmed, S. A., Alharbi, A., Hameed, A. M., & Pal, S. K. (2020). Spectroscopic studies on the biomolecular recognition of toluidine blue: Key information towards development of a non-contact, non-invasive device for oral cancer detection. Frontiers in Oncology, 10, 529132. https://doi.org/10.3389/fonc.2020.529132

4. Singh, S., Halder, A., Mohid, S. A., Bagchi, D., **Sinha, O.**, Banerjee, A., Sarkar, P. K., Bhunia, A., Ghosh, S. K., Mitra, A., & Pal, S. K. (2020). Nonthermal atmospheric plasma-induced cellular envelope damage of Staphylococcus aureus and Candida albicans biofilms: Spectroscopic and biochemical investigations. IEEE Transactions on Plasma Science, 48, 2768–2776. https://doi.org/10.1109/TPS.2020.3008282

3. Singh, S., Halder, A., **Sinha, O.**, Sarkar, P. K., Singh, P., Banerjee, A., Ahmed, S. A., Alharbi, A., Obaid, R. J., Ghosh, S. K., Mitra, A., & Pal, S. K. (2020). Nanoparticle-based 'turn-on' scattering and post-sample fluorescence for ultrasensitive detection of water pollution in wider window. PLOS ONE, 15(1), e0227584. https://doi.org/10.1371/journal.pone.0227584

Conference Proceedings

2. Jha, C. K., **Sinha, O.**, & Chakraborty, A. L. (2019). A fiber Bragg grating sensor-based wearable system to detect the pre-dicrotic and dicrotic notch in the arterial pulse pressure waveform. In 2019 IEEE 16th India Council International Conference (INDICON). IEEE. https://doi.org/10.1109/INDICON47234.2019.9030334

1. **Sinha, O.**, Singh, S., Mitra, A., Ghosh, S. K., & Raha, S. (2018). Development of a drowsy driver detection system based on EEG and IR-based eye blink detection analysis. In Advances in Communication, Devices and Networking (pp. 313–319). Springer. https://doi.org/10.1007/978-981-10-7901-6_34

INVITED SPEAKING ENGAGEMENTS

Guest Speaker · Emory School of Medicine, STORMS 2024

Guest Speaker · Penn State College of Health and Human Development Networking Expo 2022

Guest Speaker · Gaze Tracker Methods Conference, facilitated by KINARM 2022

PATENT

Co-inventor · Indian patent for an EEG-based drowsy-driver detection device 2017

PROFESSIONAL MEMBERSHIPS

American Society of Neurorehabilitation · Society for Neuroscience · National Postdoctoral Association

TEACHING EXPERIENCE

Instructor of Record Spring 2026
COS 2801 · How the Brain Learns to Move: The Science of Skill
Georgia Institute of Technology · School of Biological Sciences

Designed and independently taught a one-credit course in the Explore Living Learning Community; developed the syllabus, materials, and assessments.

Co-Instructor Summer 2025
DPT 805 · Principles of Motor Learning
Emory School of Medicine · Department of Rehabilitation Medicine

Co-taught DPT students through the Tech-to-Teaching capstone. Led laboratory activities applying motor-learning principles to practical movement problems.

Graduate Teaching Instructor 2020–2024
KINES 360 · Neurobiology of Movement
The Pennsylvania State University

Led review sessions, held office hours, developed practice questions and assessments, and supported peer study groups.

Student Mentoring

Georgia Tech and Emory Current

Research mentorship of multiple graduate students in the laboratory.

Georgia Tech ENGAGES 2025–Present

Mentored two Atlanta-area high-school students in laboratory and STEM research. Previous mentee got a full scholarship to Vanderbilt University

Emory DPT Summer 2025

Guided students in connecting movement problems with evidence-based approaches to practice and skill acquisition.

Penn State 2020–2024

Mentored two graduate students and more than six undergraduate researchers in motor control.

FELLOWSHIPS, AWARDS, AND HONORS

Jack and Dana McCallum Early Career Postdoctoral Fellowship in Neurorehabilitation	2025
Silver Medal · Penn State Graduate School Exhibition	2023
Winner · Rapid Research Contest, Penn State College of Health and Human Development	2022
University Gold Medal in Physiology · Master of Science, Presidency University	2020
Abdul Kalam Technology Innovation National Fellowship	2018
Exchange Scholar Program Scholarship · University of Groningen	2017

PROFESSIONAL AND INSTITUTIONAL SERVICE

Education Committee Member · American Society of Neurorehabilitation	2026–Present
Member · Georgia Tech Research Faculty Advisory Council	2025–Present
President · Penn State Motor Control Association	2021–2024
Judge · Penn State Graduate School Exhibition	2024

LANGUAGES

English · Bengali · Hindi